

A Guide for Trustworthy AI Implementations in the Smart Living Domain

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Abstract: Advances in AI facilitate the development of numerous smart solutions that significantly enhance efficiency and comfort for organizations and individual users. However, AI systems often suffer from the 'black box' problem and other issues that challenge their trustworthiness, and thus broad user acceptance. Since trust is crucial for the initial adoption and continued engagement with AI systems, there are numerous academic and practice driven efforts to articulate requirements for trustworthy AI applications. The sheer volume of frameworks and information can be overwhelming. This article distills prior literature and practice insights and presents a structured guide outlining the main requirements for trustworthy AI applications. While we focus on the smart living domain, our aim is to provide concrete guidance for development and reduce the complexity of the topic for SMEs seeking to develop AI applications in various application fields.

Keywords: Requirements for Trustworthy AI, Trust in AI systems, Smart Living

1 Introduction

The rapid advancements in AI and other fundamental technologies are driving the development of digital products and services that are more intelligent, complex, and autonomous than ever [Mi22]. These products and services can adapt to various contexts to better meet customers' and businesses' needs, acting as learning, constantly evolving entities [BM21]. They are increasingly integrated with other technologies and vast amounts of data, forming sophisticated digital networks that can significantly improve lives and generate unique business opportunities. However, this complexity also introduces novel ethical, legal, and social issues that can limit these systems' advantages and users' trust in such systems. For instance, the dynamic and opaque nature of these products makes tracking and understanding data collection and processing more complicated [Ro18]. Additionally, the lack of transparency in data use exacerbates existing privacy issues, with consumers increasingly demanding more transparency and control over their data [Mi22]. This is particularly the case in contexts such as smart living assistance, where such AI solutions are closely embedded in individuals' private life.

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In essence, smart living applications leverage AI and data from sensors and smart devices to provide personalized and relevant support for their user, resulting in a significant increase in efficiency, personalization and living comfort. Juxtaposed to the benefits that smart living applications afford, such AI solutions also present unique challenges that need to be addressed. One of these central challenges is building and maintaining consumer trust in smart living AI solutions. Prior research has consistently found that trust is a prerequisite for the adoption and use of technology. Thus, it is not surprising that lately we can observe increased interest in the topic of trustworthy AI from both academics and practice. While research efforts remain relatively segmented by focusing on individual challenges that AI implementations bring (e.g., explainability, fairness), or on the individual stages of AI development [Ka22] various organizations pledge openly to design and implement trustworthy AI while publishing their guidelines on the topic [Ka22]. The growing volume of insights and guidelines on how to design and implement AI applications trustworthy can lead to confusion and significant investment of effort when dealing with the trustworthy AI subject. Against this background, *the goal of this work is to present a distilled and structured guide for the main requirements of trustworthy AI applications in the smart living domain*. By transferring the principles from trustworthy AI research to the smart living domain, we want to provide concrete guidance for development and reduce the complexity of the topic for any SMEs seeking to develop AI applications in this domain.

2 Research Method

To achieve our research goal, we first perform a comprehensive literature review that entails both academic and practice papers. Defining trustworthy AI as AI that behaves as users expect it do, we first used the keywords “trustworthy AI” and “trust [and] AI”. After screening the results for their relevance (in the beginning, the title and abstract), we performed a forward and backward search of similar or referenced articles. From the resulting volume of papers, we manually extracted and structured all insights to articulate eight requirements for trustworthy AI in smart living applications. Notably, this step was critical due to (i) differing nomenclatures and (ii) different degrees of granularity within prior work. Google, for instance, commits to trustworthy AI by publishing eight AI objectives [Go24]: AI should be socially beneficial, avoid unfair bias, be built and tested for safety, be accountable to people, incorporate privacy and design principles, uphold high standards of scientific excellence, be made available to users that accord with these principles. In contrast, the EU, puts forward three broad guidelines for trustworthy AI—i.e., trustworthy AI should be lawful, ethical, and robust [EU18]. And yet, when reading the detailed description of the three EU principles one can extract further guidelines that were categorized together into the three mentioned categories [Ka22].

In addition to bringing the insights from different works to a common denominator, we conducted further literature search cycles to review prior work that focuses on only one trustworthy AI requirement (e.g., fairness, explainability) in more detail. This literature is

highly valuable since, although it provides a segmented view on the topic, it typically proposes or discusses the feasibility of technical or governance mechanisms and concrete implementation guidelines for the focal requirement. This way, we can distill concrete guidelines on how various requirements can be implemented into practice.

After having gathered and structured all relevant requirements for trustworthy AI from prior academic and practice driven sources, we discuss and transfer these insights to the smart living domain within an expert focus group. The expert group consisted of individuals working for academia and private sector companies. The research institutes mainly focus on smart service engineering and artificial intelligence (AI). The practice partners (mainly SMEs) consisted of industries consuming or providing smart living services (e.g., housing companies, assisted living service providers, Internet of Things (IoT) providers, and home automation providers).

3 Requirements for Trustworthy AI Applications

In the following we present the results of these research steps, divided into eight sections that correspond to eight requirements we deem important for trustworthy AI, and particularly, AI systems in the smart living domain.

3.1 Explainability

Explainability refers to the ability of AI systems to provide their users with understandable reasoning about decisions and outcomes [He20]. While explainability is indispensable in high stakes areas such as healthcare [On23], finance and justice systems, it is no less important in all other areas such as the smart living context. Because explainability enables users to understand AI decisions, it builds trust by making AI systems' processes transparent [Al23, Ka22]. As prior literature suggests, explainability is a prerequisite for the responsible integration of AI into society [Ry20]. It is necessary to prevent the misuse of AI and to ensure that AI systems are used in accordance with human values and ethics. Users should be able to understand how their data is processed and used. Transparent information about what data is processed where (edge vs. cloud) and for what purpose is essential when it comes to instilling user trust in such systems. Consequently, because explainability allows users to understand how inputs are transformed into outputs, which is crucial for verifying the reliability and fairness of AI, it also facilitates accountability as it enables the verification of AI decisions and identifies errors or bias in the functioning of the AI [Ka22]. While explainability is a foundational pillar for implementing other trustworthy AI requirements (e.g., fairness [Li23]), human agency and oversight, accountability or reliability, it is important to keep in mind that too many explanations can also backfire [FFL22]. The amount of explainability necessary to foster trust in AI varies depending on the context and specific use cases as well as the stakeholders involved. It is important to keep in mind that the relationship between explainability and trust is complex,

and trust also depends on other factors (e.g., perceived accuracy, fairness, and reliability) of the system [Mo23], AI system providers must engage with the target user groups to find out their needs and preferences regarding how much explainability they want. Same goes for the situations in which users require explainability and the communication forms they wish to receive it. After all, prior research suggests that the level of explainability and the way it should be articulated will also depend on heterogeneous user preferences and characteristics (e.g., users' familiarity with AI [Tu22]).

Taken together, in the smart living context, explainability should focus on clarity and simplicity. Users are very heterogeneous regarding their technical expertise and generally do not require detailed technical explanations of the underlying AI algorithms of smart living applications. Instead, they wish practical solutions that simply fulfill their task. However, users of such services need to understand what the AI does and why it performs certain actions [Tu22]. The smart living domain is particularly special here, since users interact with an AI system in various situations throughout the day. To ensure that users understand AI systems' reasoning and feel comfortable with the technology—especially with varying levels of cognitive load—explainability has to be adaptable to the users' needs and preferences. For instance, AI systems could initially provide a basic explanation to all users, with the option to retrieve more detailed information if the users wish to do so [Ga20]. This approach respects the different needs and level of curiosity of different user groups and makes the technology more accessible and trustworthy.

3.2 Accountability

Accountability envisions that AI developers and operators can be held responsible for their systems' behavior and outcomes. Achieving accountability in AI requires implementing mechanisms that allow claims about AI systems to be verified [Ka22]. This can be achieved through appropriate testing, audits, and suitable monitoring frameworks [Ka22]. As mentioned previously, accountability is innately linked to explainability. Only if AI systems have a certain level of transparency and one can trace and understand how inputs lead to a certain output, one can assign responsibility to those involved in a particular part of the system. Although developers are one obvious stakeholder group that can be held responsible for an AI components or system's erroneous outcome, scholars increasingly suggest that accountability systems should be integrated throughout the AI lifecycle [Ja18], rather than punctually (e.g., in the AI development phase). Furthermore, as AI solutions grow more complex by combining multiple technologies (e.g., IoT) with various AI components, accountability needs to be ensured not only on the individual component level but also on an aggregate 'solution' level, where components work together to provide a wholistic service to their user. To underline this paradigm, [MNI21] proposed a Global-view Accountability Framework (GAF) that ensures accountability in environments where multiple AI systems interact with each other.

For the smart living domain, prior literature proposes several strategies to ensure that devices and systems within the smart home environment are traceable and thus responsible

parties can be made accountable when necessary. For instance, a smart alarm system might utilize sensor information from another system, such as an ambient assisted living solution. Accountability should govern who or what system is responsible for a particular action (e.g., if there is no alarm in case of a burglary). One potential option is to implement so-called *witness-based accountability systems* [MX16]. This method involves using IoT devices mutual verification entities that can confirm the behavior of another smart device or AI system. This way, this method helps create a system where devices can demonstrate the correctness of their actions to other devices, thereby increasing accountability. Similarly, *accountable communication protocols* [LXG13] use a peer-review strategy, where devices verify and confirm each other's data. The implementation of a *data transparency and user control mechanism* [Ja18] is another way of enabling accountability. By making the input-output process transparent, users can understand and override any erroneous AI system decisions and call upon the entities that caused erroneous outcomes.

3.3 Fairness

The fairness requirement envisions that AI systems operate without bias and treat all users equitably. It refers to the principle that AI systems should make decisions impartially, justly, and without prejudice, especially regarding sensitive attributes such as race, gender, or age. Achieving fairness in AI is crucial to ensuring that technology benefits all segments of society without discrimination or exclusion [Al23, Ka22]. Researchers in this field agree on its importance and point out that there are various types of biases that can affect AI systems, such as selection bias, measurement bias, and algorithmic bias [Ka22]. One of the primary challenges in ensuring fairness involves dealing with biases in training data [Ka22]. Thus, it is not surprising that extant literature proposes a variety of techniques to address this challenge. Some examples include: a reweighting of training examples or modifying learning algorithms to penalize fairness violations. However, because biases can also stem from the way the systems are developed and deployed, AI systems must undergo rigorous testing and validation to identify and address any potential biases [Al23]. Thereby, testing and validation must be understood as an ongoing process rather than a once-off activity. This is particularly important when considering that AI systems evolve over time by learning from new data and users' inputs. Consequently, integrating feedback mechanisms and adjusting AI systems based on audit results are crucial for the ongoing assurance of fairness. Besides these strategies, fairness can also be partially achieved by ensuring that AI is developed by diverse teams. Diverse teams are more likely to recognize and address potential biases in AI systems, leading to fair outcomes [Ro23].

In the smart living context, fairness plays a pivotal role since AI systems interact with a very diverse set of users belonging to different genders, ages, socio-economic groups, cultures, and races. These systems must perform equally good for all user groups and should account for their varying preferences. For instance, a smart camera granting access to an apartment should work independent of the inhabitant's age, gender, or race. Furthermore, they should allow for personalization, since users must always balance

comfort, cost, and value [So20]. Accordingly, to achieve a high degree of fairness from all point of view (technical and socio-economic), smart living solution providers must engage closely with their target user groups and elicit their needs to try achieving these needs without prejudice and bias-free.

3.4 Data Privacy

Another key requirement for trustworthy AI systems is related to data privacy. For users to trust complex AI systems, developers and providers must ensure that data collection, storage, and usage comply with relevant privacy laws, ethical standards and user's willingness to provide and accept the processing of their data. While privacy is fundamental for trustworthy AI, it is also essential to balance protecting privacy and promoting technological innovation. Regulatory frameworks such as the EU AI Act [EU24] aims to achieve this balance by mandating the integration of various privacy principles throughout the lifecycle of AI systems. Exemplary principles are privacy by design, transparency and information obligations, or protection of special data types. Yet, besides regulatory frameworks, there are also various technical measures that can be taken to integrate privacy into AI systems. [JG20] for instance, examine privacy-preserving techniques essential for training AI models without compromising user privacy. These techniques are crucial due to the large data volumes required for AI operations, inherently posing a privacy threat. [Ca20] explores using privacy technologies and cryptographic methods to protect data within AI inference systems, ensuring that customer data is protected throughout the AI lifecycle. On another note, [Ku21] introduce an information-theoretic framework for trustworthy AI, incorporating privacy leakage measures and demonstrating how privacy aspects can be optimized during system development.

In the smart living context, where smart living solutions do not only entail AI technologies but also many IoT devices, there is a need for a holistic approach to privacy. Sensors are potentially ubiquitous in the personal spaces of smart living inhabitants and can sense very sensitive information. Thus, smart living services need to evaluate which kind of data is satisfactory to reach a task —under the premise of data minimization. Besides, adjusting the data resolution and volume can benefit data privacy in the domain [KL24]. Furthermore, it is essential to approach privacy with a comprehensive strategy encompassing technical security measures, ethical data processing practices, and compliance with legal standards.

3.5 Security and Robustness

While privacy mainly focuses on how to minimize the extent of sensitive data a system collects and processes, security focuses on protecting the data and the functionality against unauthorized access. To protect AI systems from potential cyber threats there is a need for dedicated cybersecurity measures [Al23, Ka22], which in turn requires specialized expertise and technologies. One criterion connected to security is robustness. Robustness

refers to the requirement that AI systems must operate accurately and reliably under a variety of conditions, including challenging or new, unexpected scenarios. This includes the ability of AI systems to handle noisy data and recover from (partial) failures without significant performance degradation [Al23, Ka22]. There are various ways to achieve robustness of AI systems. However, the most promising path seems to be a security-aware strategy combining rigorous testing, continuous monitoring, and iterative improvement based on feedback from real-world use. Potential vulnerabilities need to be addressed throughout the AI lifecycle (e.g., during both the training and inference phases of AI systems). A systematic lifecycle approach to robustness that considers all stages from data collection and modeling to deployment and continuous monitoring, ensures that security and robustness are integrated at each stage, with specific strategies aimed at addressing the unique challenges at each point. After all, similar to other requirements in this work, ensuring security and robustness is not a once-off matter but rather a continuous task that needs to be monitored and adapted over time [Li23]. Extant literature discusses various architectural and training strategies that can enhance AI systems. For instance, it is important to achieve strength against inference attacks, where attacker wants to gain insights in the AI model through strategic input and queries, and poisoning attacks, where the attacker intentionally injects faulty or manipulated data into the training dataset of an AI model. Protecting against such attacks requires architectural and training choices that increase the AI system's ability to tolerate irregular queries and malicious manipulations without performance degradation.

In the context of smart living robustness needs to account for several unique solution characteristics that might not necessarily apply to other settings: Firstly, smart living systems must *operate in a non-intrusive manner* for the residents. Thus, robust AI systems should function reliably under various conditions without needing constant adjustments or interactions from the user. Furthermore, *AI systems operate in real-time and must deal with diverse and dynamic environments*. Hence, error detection must be adaptive, in real-time. Given the critical nature of services provided by smart homes, such as those related to health and safety, robustness includes the ability to adaptively detect and recover from errors—i.e., adapt to changes in environment to maintain functionality and provide accurate support even in previously unknown situations. Finally, the smart living software typically runs on local edge devices in the home of inhabitants. These systems must rely on regular updates, preferably automatically, to counteract new and upcoming threats.

3.6 Human Agency and Oversight

Human agency and oversight emphasize the importance of keeping AI technologies under human control and ensuring that decisions made by AI systems can be monitored, and—if necessary—corrected by humans [Al23, Ka22]. This requirement ensures that AI supports human decision-making rather than replacing it altogether. This is important against the background that AI systems grow increasingly autonomous in their decision-making process [BM21]. Enforcing human agency and oversight can be achieved by designing AI systems with clear protocols for human intervention and establishing

guidelines on how and when humans should interact with AI. Simultaneously, there is a need to educate users about their options to exert control and override AI systems' decisions. If users are unaware that they can and should review and override AI decisions, they will not do so. As part of such user-related educational campaigns, AI system providers can involve their target user groups in the development process of AI interfaces. An intuitive and user-friendly interface is crucial to ensure that all users, regardless of their technical knowledge, can use the system effectively. This reduces frustrations but also ensures fairness, and ultimately trust in such systems.

In the context of smart living applications, we envision a balanced approach that incorporates human agency and oversight while making the most of technological advancements. We fully acknowledge the importance being able to manually override AI systems' decisions, and the value of human input for providing more personalized smart living assistance. However, we also feel compelled to highlight that frequent manual interventions can impair the efficiency and effectiveness of AI systems. Hence, the goal is to find the sweet spot where human interactions with AI systems are either necessary or encouraged. If human oversight and input is necessary, fairness issues might arise, with less technically versed user groups being unable to enjoy the full range of benefits smart living systems could afford. And yet, for those who seek control AI systems must allow users to exert the level of control they desire.

3.7 Reproducibility

Finally, reproducibility refers to the ability to consistently produce the same outcomes under the same conditions [AI23]. It also involves applying principles that ensure AI artifacts can be reliably replicated and verified in real-world applications. Prior literature suggests various strategies to achieve reproducibility. *Documentation of AI methods and data*, for instance, can propagate the information needed for independent reproduction [GGA18]. Another way is sharing code and data with other entities that can independently reproduce the outcomes of the AI system. This *open source and open data* approach not only facilitates verification but also promotes collaboration and innovation in the field [GGA18]. In a similar vein, using *standardized tools and environments* might help achieve reproducibility. After all, the use of such tools can help standardize AI development processes and make result reproduction easier [GGA18]. Following industry *best practices and checklists* could yield high levels of reproducibility [KT19]. These exemplary strategies can be complemented by *educational campaigns for AI practitioners and community building activities*. While educating AI practitioners about reproducibility and its importance can foster a culture of transparency and rigor in AI development, building communities that are aware of the reproducibility issues across the entire AI lifecycle entice AI developers to share tools, and practices that help reproducibility efforts.

In the context of smart living applications, reproducibility can be influenced by various factors, including the consistency of technology performance and the reliability of the involved IoT devices and AI systems. Because smart living applications are typically

characterized by high number of IoT devices and AI systems orchestrated to provide a service, it is crucial to ensure not only the reproducibility of outcomes on the individual component level but also of the entire smart living solution as an ensemble [Ra16]. A particular problem in this domain is that the input data of algorithms is typically high-dimensional, making it hard to replicate a specific situation repeatedly. Thus, AI algorithms must abstract the situational context from the specific sensor information first.

4 Discussion

The goal of this work was to provide a compiled overview over the key requirements for designing and implementing trustworthy AI. Based on a comprehensive literature review and a refinement focus groups with SME representants, we discussed seven requirements for trustworthy AI applications in the smart living domain. One of the foundational requirements which is partly a prerequisite for other requirements is *explainability*. Explainability addresses the so called ‘black box’ problem of AI technologies and implies that AI systems must be able to provide understandable reasoning to humans for decisions and outcomes. For smart living applications the best level of explainability is one that (i) prioritizes clarity and ease of use, (ii) providing explanations that are directly relevant to the tasks the AI is performing and (iii) are adaptable to the user's level of interest and expertise. Another requirement for trustworthy AI is *accountability*. Innately linked to explainability, accountability involves ensuring that there is a clear and enforceable mechanism to hold designers, operators, and providers of AI systems accountable for their systems’ actions and decisions. This requires the ability to trace decisions back to the entities that made them (i.e., explainability), ensuring that AI systems are *fair* to all users (i.e., just and equitable). Addressing fairness involves technical solutions to mitigate biases, adherence to ethical and legal standards, and transparency in AI operations. Mechanisms to detect, address, and prevent biases in AI systems must be implemented, including the use of diverse and representative training data, the development of algorithms that can identify and correct biases, and continuous monitoring to ensure that AI systems continue to operate fairly as they are scaled and deployed in various contexts. Another key requirement is *privacy*, which is crucial to maintaining user trust, complying with legal standards, and promoting the responsible development and deployment of AI technologies. Furthermore, trustworthy AI needs to be *robust* and *secure*. While robustness involves developing systems that are effective not only under ideal conditions but also capable of maintaining their functionality and integrity under less-than-ideal or actively hostile conditions, security involves ensuring that AI systems are protected against external threats that could disrupt their operation or lead to the misuse of the data they process. Security also implies that systems’ decisions cannot be ‘hijacked’ by attackers and *human agency and oversight* is always possible. Despite the autonomy and capabilities of AI systems, humans ultimately remain responsible for significant decisions and actions. Thus, they should also be the ones able to override and correct AI systems’ decisions at any time they feel it is necessary to do so. This requirement protects against the risks of uncontrolled AI autonomy and ensures that AI technologies are used

responsibly and ethically, in alignment with human values and societal norms. Ultimately, *reproducibility* implies that decisions made by AI should be reproducible under the same conditions to ensure consistency and reliability in the results.

Notably these requirements are interconnected and should be approached from a holistic perspective that covers the entire AI lifecycle. Due to the foundational nature of explainability, which is a prerequisite for other requirements one might be tempted to focus on explainability only. This, however, will not lead to the desired levels of trust in AI. Because trust is a composite concept that depends on many variables, trustworthy AI will require a comprehensive strategy that addresses all requirements to a satisfactory level across all AI lifecycle phases. While once-off activities to comply with the requirements listed in this work might ensure that AI systems are trustworthy at the beginning of their implementation, their agentic nature that allows them to learn and evolve in time requires systematic and repeated measures over time. Such measures might include monitoring consumer sentiment and trust levels, and then pursuing transparent communication strategies regarding functional and security updates or implemented customer feedback. After all, as prior literature on trust in IT systems suggests, individuals' initial trust can develop over time in accordance with the experiences they have made with these systems. While the initial trust is particularly crucial for adopting and initially interacting with technology, trust levels during the post-adoption phase determine users' engagement and effective use of systems. Consequently, building and maintaining trust in AI systems for smart living not only ensures initial adoption of such systems but also fosters long-term engagement, ultimately paving the way for a future where technology seamlessly enhances our everyday lives.

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